Chenshuhao (Cody) Qin

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Education

University of Southern California (USC) — Los Angeles, CA
Master of Science in Computer Science — *Jan. 2026 – Dec. 2027 (Expected)*

- **Coursework**: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Predictive Modeling

Duke University; Duke Kunshan University — Durham, NC / Kunshan, China
Bachelor of Science in Computer Science; B.S. in Computation and Design — *Aug. 2021 – May. 2025*

- **Honors**: **Apple WWDC23 Swift Student Challenge Winner** (1 of 13 in China), Provincial Innovation Grant Recipient ($3k), DKU Student Ambassador

Skills

Languages: Python, C, Java, SQL, TypeScript/JavaScript, Swift
Big Data & Lakehouse: Apache Spark, Flink, Iceberg, Kafka, Ray, batch & streaming pipelines, object storage (OSS/S3)
Infrastructure & Cloud: Kubernetes (incl. Operator development), Docker, Linux, Git, CI/CD, MySQL, PostgreSQL, Redis, REST APIs (FastAPI, Flask), React, AWS
AI & Machine Learning: PyTorch, TensorFlow, CUDA GPU programming, LoRA fine-tuning, vLLM, Hugging Face, Pandas, NumPy
LLM & Agents: agent harness & context engineering, retrieval-augmented generation (RAG), LangChain, LlamaIndex, AgentScope, Milvus vector search

Work Experience

Alibaba Cloud (Top-4 Global Cloud Provider) — Hangzhou, China
AI Engineer Intern (Agent Infrastructure) — *Jun. 2026 – Aug. 2026*

- Collaborated with Alibaba's **Tongyi Lab** to productize **Data-Juicer Agents** — an open-source **agent harness** — into natural-language multimodal data pipelines (cleaning, filtering, deduplication) for the broader **Qwen** ecosystem, deployed on **ADL Foundry**, the team's Apsara data-lake analytics platform.
- Built internal-platform integrations coupling **Spark**, **Flink**, and **Apache Iceberg** lakehouse engines with Data-Juicer Agents, driving **batch** and **Kafka**-backed **streaming** pipelines that turn raw multimodal data into analysis-ready datasets on **OSS** object storage.
- Built an agent-to-**Ray** distributed execution pipeline over private-cloud object storage (OSS) and **Kubernetes**, with hands-on **Kubernetes Operator** development for job lifecycle automation — **2,793 records, 16 operators, ∼5 minutes** — landed upstream through a merged PR.
- Analyzed **agent execution traces** to localize context bloat; designed a three-layer context-control architecture (tool-output compaction, memory compression, hard token budgets) cutting single-tool context payloads by **82%** and stabilizing multi-turn agents on 30k-token models.
- Shipped a full-stack agent workspace (**React / FastAPI / MySQL**) with persistent sessions, execution tracing, conversation rollback, and multi-instance isolation (**4 container-image releases**); contributed **5 merged PRs** (unit-tested, CI green) to the 7k⋆ Data-Juicer ecosystem, plus 1 proposal adopted by maintainers.

Blueberry AI — Burlingame, CA (Remote)
AI Engineer Intern — *Nov. 2025 – Jan. 2026*

Built a **Milvus + MySQL** feature-extraction and auto-tagging pipeline batch-indexing **200K+ images** for landmark / IP / SKU retrieval; benchmarked **DINOv3, Qwen-VL, and BGE-VL** under a unified recall / precision / latency framework; delivered a YOLO + vector-search retrieval POC validated for commercial use.

Tsinghua Sichuan Energy Internet Research Institute — Chengdu, China
Algorithm Engineer Intern — *Sep. 2025 – Oct. 2025*

Engineered a **DeepLabV3+ (PyTorch)** segmentation model for microgrid fault diagnosis reaching **92% accuracy**; owned the full data lifecycle of **500+ images** from preprocessing to deployment, anchoring winning infrastructure bids.

Leiting Games (G-bits) — Shenzhen, China
AI R&D Intern (AIGC & Automation) — *May. 2025 – Aug. 2025*

Built a **Dify + RPA** automated monitoring pipeline (**98.85% accuracy**, 1,000+ violations detected) and fine-tuned **SDXL / Flux LoRA** models driving **117K+ installs, ¥24.6M billings (196% ROI)**; led a 5-person team to **3**× creative output (2025 Best Practice).

Projects & Publications

ChatDKU — RAG Campus Q&A Assistant | *Python, LlamaIndex, LangChain, Redis* — Jul. 2024 – May. 2025

Co-founded a university-adopted **RAG** Q&A agent indexing **12,000+ webpages** over a privately deployed LLM; implemented **hybrid search (BM25 + vectors)**, **HyDE**, multi-level **Redis** caching, and **SSO** web deployment; officially deployed at DKU and documented as a DKU Signature Work.

Publication: *Privacy-Preserving Room Occupancy Estimation Using Federated Analytics of BLE Packets* — **ACM SenSys 2024** (Demo Track, second author), pp. 889–890; **95%** real-world accuracy via federated learning over BLE.